VitalSync: A Decoupled IoMT Architecture for Risk Triage and Resilient, Faithfulness-Verified Clinical Summarization

Bhavansh Goyal

***Abstract*—Continuous monitoring in the Internet of Medical Things produces physiological telemetry far faster than clinicians can read it. Generative summarization promises relief, but it introduces a component that is slow, intermittently unavailable, and capable of stating things the data does not contain. Placing such a component in the ingestion path means inheriting its worst case. We present VitalSync, an architecture that separates three concerns prior systems couple: telemetry ingestion, deterministic risk triage, and generative summarization. We make four contributions. First, we audit a widely downloaded public vital-signs dataset and recover its label-generating rule exactly: a three-clause Boolean over heart rate and body mass index reproduces all 200,020 published labels without a single mismatch. Classifiers reach the ceiling by recovering that Boolean, and an ablation restricted to the remaining vital signs performs below the majority-class baseline, so reported accuracies on this dataset are artifacts of label construction. Second, we separate latent physiology from sensor observation and show that a learned triage model recovers the clean-signal NEWS2 escalation band more reliably than the NEWS2 rule applied to the same corrupted readings, with the advantage growing monotonically in sensor noise and reversing at zero noise, where the rule is exact by construction. That reversal is the negative control that makes the rest credible. Third, we show that macro-F1 selects a model reporting high-risk observations as low risk 5.9 times more often than a cost-aware criterion, and we report the disagreement rather than resolving it after the fact. Fourth, we specify a summarization cascade in which a mechanical numeric-entailment check governs fallback, terminating in a deterministic generator that cannot fail. The implementation sustains 100 requests per second at 27–31 ms p95 ingestion latency, returns in under one second while the triage service is stalled for three, and continues to ingest and summarize with that service stopped altogether. All results are software engineering evaluation on simulated cohorts; no clinical claim is made.**

***Index Terms*—Internet of Medical Things, remote patient monitoring, early warning score, clinical decision support, large language models, hallucination detection, dataset auditing, fault tolerance, system resilience, model calibration.**

I. Introduction

REMOTE patient monitoring generates continuous streams of vital-sign telemetry. The limiting factor is not acquisition. It is attention. A ward producing thousands of readings an hour presents clinicians with far more data than anyone can review, and the conventional response—one alarm per parameter per threshold breach—scales with the product of patients, parameters, and sampling rate. Studies of physiologic monitors in intensive care have counted more than two and a half million alarms in a single month on one unit [6], with the large majority non-actionable [5]. The result is desensitization, and desensitization has been linked to patient deaths.

Large language models look like an attractive remedy. They turn columns of numbers into prose a clinician can read in seconds, and recent work reports that adapted models can match or exceed medical experts on several clinical summarization tasks [29]. But they also carry properties a monitoring system should not inherit. Latency is measured in seconds rather than milliseconds. Availability depends on a remote service that may be rate-limited or down. And the models can produce fluent text asserting values that never appeared in the input—a failure mode documented across summarization generally [28] and surveyed at length for natural language generation as a whole [27]. A system that puts such a component in the path of telemetry ingestion inherits all three.

VitalSync separates these concerns. Ingestion terminates at a durable write. Risk triage is a deterministic, millisecond-scale operation that runs independently. Generative summarization sits behind a cascade whose final tier is a template generator with no external dependency. The architectural claim is availability: no single language model is required for the system to function, and no unverified number reaches a reader from a non-final tier.

There is a second thread to this paper, and it was not planned. The project was specified around a public dataset. Before training we audited the label, which is cheap and rarely done. What we found reframed the work: the label is a three-clause rule over two variables, the dataset contains no abnormal reading at all, and the high accuracies reported on it measure Boolean recovery rather than clinical prediction. We report that as a primary contribution rather than working around it, because the dataset is public, widely downloaded, and the defect is invisible to the standard train-and-report workflow.

A. Contributions

1) A label-provenance audit of a public vital-signs dataset that recovers its exact generating rule and demonstrates, by ablation, that the vital signs it distributes carry no information about its published label (Section IV).
2) An experimental design that separates latent physiology from sensor observation, turning "learned model versus threshold rule" into a falsifiable comparison, together with the zero-noise negative control that validates it (Section V).

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This work describes a research prototype. The system is not a medical device and has not been evaluated for clinical use. All cohorts are simulated; no patient data was used at any stage.

3) Evidence that aggregate classification metrics select triage models that are materially worse under clinically asymmetric cost, quantified on both a standard cohort and a reduced sensor-only deployment (Sections V and IX).
4) A summarization cascade in which mechanical numeric entailment governs fallback, with a deterministic terminal tier that guarantees a response (Section VII), and an implementation whose decoupling, throughput, and degraded-mode behaviour are measured rather than asserted (Section IX).

II. Related Work and Background

A. Early warning scores

Aggregate early warning scores condense several vital signs into a single number with defined escalation bands. The Modified Early Warning Score [3] established the pattern; the National Early Warning Score, standardized by the Royal College of Physicians and revised as NEWS2 [1], is now mandated across the NHS. Its discrimination has been measured on large observational cohorts: against the combined outcome of cardiac arrest, unanticipated ICU admission, or death within 24 hours, NEWS achieved an area under the receiver operating characteristic curve of 0.873, outperforming 33 competing scores [2].

Scores of this kind have two properties that matter here. They are deterministic, so a clinician can recompute one by hand from the chart. And they are additive over parameters, so the total decomposes into contributions that explain it. A systematic review of 95 early warning score papers found that every included study was at high risk of bias and that many scores in daily clinical use have real methodological weaknesses [4], which is a caution against treating any such score as ground truth rather than as a documented, auditable convention. We use NEWS2 in exactly that second sense.

B. Machine learning for clinical deterioration

A large literature replaces or augments these scores with learned models. Churpek *et al.* compared machine learning methods against conventional regression for ward deterioration across multiple centres [8]. Deep models have been applied to continuous prediction of acute kidney injury at national scale [10] and to early prediction of circulatory failure in intensive care [11]; sepsis prediction in the ICU has its own systematic review [12]. Deployed at scale, automated deterioration identification coupled to rapid-response workflows has been associated with reduced mortality [9]. The broader promise and the broader caution are both well surveyed [13].

The caution matters. External validation of a widely deployed proprietary sepsis model found poor discrimination and calibration, with many onsets missed [7]. Dataset shift between development and deployment is a recognised clinical hazard rather than a theoretical one [14], and the assumption that a model validated in one setting transfers to another has been argued to be a myth [15]. None of these failures is visible from an aggregate accuracy figure on a held-out split.

C. Alarm burden

Alarm fatigue is the operational problem a triage layer is meant to address. Observational work in intensive care documents both the raw volume and the low actionable fraction [6], and the patient-safety framing is established [5]. We are careful in this paper to distinguish counting notification events, which we do, from measuring clinician workload or fatigue, which we do not.

D. Language models over clinical text

Clinical summarization with language models is an active area, with reported results that are genuinely strong under expert review [29]. Faithfulness is the open problem. Systematic review of the area finds generative models highly prone to producing content unfaithful to the source, and reports that entailment-based measures track faithfulness more closely than standard overlap metrics do [28]. The general survey of hallucination in natural language generation catalogues both the taxonomy and the mitigations [27]. Retrieval augmentation reduces but does not eliminate the problem [30].

Our contribution here is narrow and deliberately so. We do not attempt to make a model faithful. We check its output mechanically against the numbers it was given, and we use the result of that check to decide which tier answers.

E. Machine learning systems engineering

The engineering literature on deploying machine learning is directly relevant to an architecture paper. The operational practices collected under MLOps [33] and a survey of deployment case studies [34] both identify the glue around the model—serving, monitoring, data dependencies—as the dominant source of risk. Tail latency behaviour in distributed services [32] motivates the bounded-executor and timeout design we adopt: a component that is occasionally slow degrades a coupled system far more than its average latency suggests.

F. Dataset quality, leakage, and interoperability

Leakage is a widespread failure mode in machine-learning-based science, affecting hundreds of papers across seventeen disciplines, with a taxonomy of eight distinct types [17]. In medical imaging specifically, a review of COVID-19 models found none fit for clinical use, with dataset problems prominent among the causes [16]. Public critical care datasets such as MIMIC-III [18] have raised the standard for provenance by documenting their construction. The dataset we audit in Section IV does not meet that standard, and the audit is in the same spirit as this body of work. For data exchange we follow HL7 FHIR, whose RESTful resource model [31] shapes the observation contract used throughout.

III. Problem Formulation and Requirements

A. System model

Let a telemetry source emit observations indexed by time. Each observation is a vector

$$x = (x_{\mathrm{hr}}, x_{\mathrm{spo2}}, x_{\mathrm{temp}}, x_{\mathrm{rr}}, x_{\mathrm{sbp}}, x_{\mathrm{dbp}}) \tag{1}$$

of six sensed channels, optionally extended with glucose and with two staff-recorded context flags: supplemental oxygen $o \in \{0, 1\}$ and consciousness $a \in \{0, 1\}$ on the AVPU scale. Derived quantities are computed on write:

$$\mathrm{SI} = x_{\mathrm{hr}}/x_{\mathrm{sbp}}, \quad x_{\mathrm{sbp}} > 0 \tag{2}$$

$$\mathrm{PP} = x_{\mathrm{sbp}} - x_{\mathrm{dbp}} \tag{3}$$

$$\mathrm{MAP} = (x_{\mathrm{sbp}} + 2x_{\mathrm{dbp}})/3. \tag{4}$$

A triage function f maps an observation to an escalation band $\hat{y} \in \mathcal{B} = \{\textsc{low}, \textsc{medium}, \textsc{high}\}$ with a probability vector $p \in \Delta^2$. A summarization function g maps an observation, a bounded trend window, and the triage output to a short text.

B. Failure model

We assume the following may occur independently and at any time, and the system must remain useful under each.

- **F1, triage unavailable.** The service implementing f is unreachable, returns an error, or exceeds its timeout.
- **F2, generator unavailable.** A tier implementing g is unreachable, rate-limited, or exceeds its timeout.
- **F3, generator unfaithful.** A tier returns well-formed text that asserts a value absent from its inputs, or names a condition.
- **F4, overload.** Offered telemetry exceeds the rate at which analysis can complete.
- **F5, incomplete observation.** One or more channels are missing, or the staff-recorded flags are unavailable because the feed is sensor-only.

F3 is the failure that motivates most of this paper, because unlike the others it is silent. A tier that is down announces itself. A tier that invents a plausible lactate value does not.

C. Requirements

- R1 **Ingestion independence.** Acknowledgement latency must not depend on the latency of f or g.
- R2 **Bounded resource use.** Under F4 the system must apply backpressure rather than accumulate unbounded work.
- R3 **Honest degradation.** Under F1 the band must be reported as unavailable, never inferred.
- R4 **Total summarization.** Under F2 and F3 a summary must still be produced.
- R5 **Grounded output.** No text from a non-final tier may assert a value absent from its inputs.
- R6 **Auditability.** Every assessment must record which tier answered, whether triage was reachable, and whether verification passed.

Section IX evaluates each of these against the implementation.

IV. Track A: Auditing the Dataset

A. Motivation

The project was specified around a public Kaggle dataset of human vital signs [37] (200,020 rows, 17 columns) carrying a `Risk Category` label, to be classified with random forest and gradient boosting. Before training we audited the label. This is cheap, and given the documented prevalence of leakage [17] and dataset-driven failure [16], it should be routine.

TABLE I
Track A. Classification on the public vital-signs dataset, and the recovered label rule. Scores approach the ceiling because the published label is a deterministic function of heart rate and BMI, not because vital signs predict risk.

Model	Accuracy	Macro-F1
random_forest	1.00000	1.00000
gradient_boosting	1.00000	1.00000
Recovered rule	1.00000	–
Vitals-only ablation	0.51850	0.42085
Majority-class baseline	0.52552	–

Rule: `(Heart Rate > 90) OR (Derived_BMI > 30) OR (Derived_BMI < 18.5)`, exact on 200,020 of 200,020 rows (0 mismatches). The ablation uses only SpO_2, blood pressure, temperature and respiratory rate, and does not beat the majority-class baseline.

B. Method

We computed per-class descriptive statistics for every numeric column, ranked features by mutual information with the label, fitted a depth-3 decision tree and exported its rule, tested a candidate Boolean against all rows, reproduced the specified random forest and gradient boosting experiment, and ran an ablation restricted to the vital signs that showed no class separation.

C. Result: the label is a rule

The published label is reproduced exactly, with zero mismatches across all 200,020 rows, by

$$\mathrm{High} \iff (\mathrm{HR} > 90) \vee (\mathrm{BMI} > 30) \vee (\mathrm{BMI} < 18.5). \tag{5}$$

Three consequences follow, in increasing order of damage.

The dataset is binary. There is no NORMAL class, so the three-class target originally specified cannot be trained from it at all.

Every vital sign lies inside its normal reference range: heart rate 60–99, oxygen saturation 95.0–100.0, systolic pressure 110–139, temperature 36.0–37.5, respiratory rate 12–19. The cohort contains no physiologically abnormal observation. It is a dataset of healthy people with a label attached.

Class-conditional means for oxygen saturation, both blood pressures, temperature, respiratory rate, and mean arterial pressure agree to three decimal places. Those variables carry no label information whatsoever.

Table I and Fig. 1 report the classification results alongside the rule and the ablation. Both ensembles reach the ceiling. The ablation, trained only on oxygen saturation, blood pressure, temperature, and respiratory rate, does not beat the majority-class baseline.

D. Implication

A classification score on this dataset measures recovery of Equation 5. It is not evidence of clinical predictive capability. A reviewer fitting a shallow tree finds this in seconds, which makes publishing such a score a reputational risk as much as a scientific one. We therefore do not use this dataset to train the deployed model, and we recommend that any classification result on a public physiological dataset be accompanied by

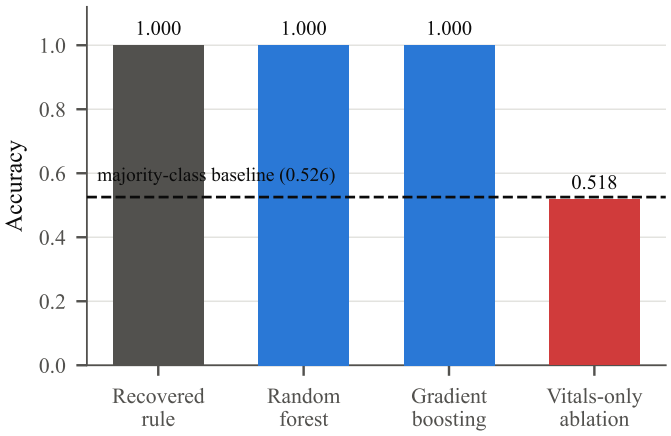


Fig. 1. Track A. Both ensembles reach the ceiling by recovering the label rule. The ablation trained only on oxygen saturation, blood pressure, temperature and respiratory rate falls *below* the majority-class baseline: those channels carry no information about the published label.

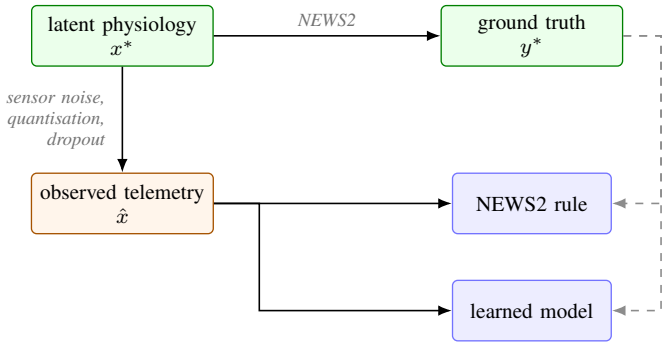


Fig. 2. Separating latent physiology from sensor observation. Ground truth is scored on the latent signal; every method, including the rule baseline, observes only the corrupted telemetry; the dashed edges indicate that both are scored against the same latent-signal truth. This is what makes "learned model versus threshold rule" a real comparison rather than a restatement of the labelling function.

evidence that the label is not a deterministic function of the features.

V. Track B: Triage Under Sensor Noise

A. The circularity problem

Substituting NEWS2 for the dataset's label does not by itself solve anything. NEWS2 is also a deterministic function of vital signs, so a model trained to reproduce NEWS2 from clean vitals is learning a rule that is already known, and any reported accuracy is a statement about function approximation rather than about clinical utility. Swapping one rule for another changes which tautology is being measured, not whether one is.

B. Design: latent truth versus observation

We break the circularity by distinguishing what is true from what is measured:

$$x^* \xrightarrow{\text{NEWS2}} y^*, \qquad x^* \xrightarrow{\text{sensor noise}} \hat{x},$$

where x^* is latent physiology, y^* the ground-truth escalation band, and $\hat{x}$ the corrupted telemetry. Ground truth is scored on x^*. Every method—including the rule baseline—observes only $\hat{x}$. The rule baseline computes NEWS2 on $\hat{x}$, which is precisely what a conventional threshold monitor does in practice. The learned model is trained on $(\hat{x}, y^*)$ and can therefore internalize the noise distribution, soften band boundaries, and exploit cross-parameter redundancy that a per-parameter rule cannot.

This yields a falsifiable comparison with a built-in negative control: at zero noise $\hat{x} = x^*$, the rule is exact by construction, and the learned model *must not* beat it. Fig. 2 shows the arrangement.

C. Cohort and corruption model

Latent severity $z \in [0, 1]$ is drawn from a three-component mixture: 60% stable, 25% mild derangement, 15% acute deterioration, with Beta shapes keeping each component inside its own region. Four deterioration archetypes (respiratory, circulatory, febrile, mixed) weight the channels differently, so that deterioration appears across several parameters at once rather than in one channel independently. This matters: a generator with independent marginals would make the aggregate score trivially redundant and would flatter any method that exploits aggregation.

Corruption is applied per channel: Gaussian noise on all channels, motion artifact on heart rate at 1.5% of readings, perfusion dropout on oxygen saturation at 2%, display quantization, and 3% whole-channel dropout imputed at inference. Quantization is treated as part of the corruption and is disabled at zero noise, so that the zero-noise condition is an exact identity and can serve as a test of the generator and scorer themselves. Appendix B gives the parameters.

D. Metrics

All estimators are implemented with scikit-learn [22]. The random forest [19] and gradient boosting [20] configurations follow the original project specification; the gradient boosting family itself is reviewed in [21].

Macro-F1 is the pre-registered selection criterion. It is insufficient alone: it scores a missed high-risk observation identically to a spurious one. We therefore also report high-risk recall, *missed-HIGH* (the fraction of high-risk observations reported as low risk), and a mean cost under an asymmetric matrix penalizing under-triage far more heavily than over-triage, following established cost-sensitive practice for imbalanced problems [24], [25]. Those weights are a stated assumption, not derived from outcome data; Fig. 3 shows the matrix and Appendix C states it explicitly.

Calibration is assessed by expected calibration error over ten equal-width bins and by the multiclass Brier score, following standard practice [23]. A confidence value displayed to a clinician that does not mean what it says is worse than displaying none.

TABLE II
TRACK B. RECOVERING THE CLEAN-SIGNAL NEWS2 BAND FROM SENSOR-CORRUPTED TELEMETRY. ALL ROWS SEE THE SAME CORRUPTED OBSERVATIONS. MISSED-HIGH IS THE FRACTION OF HIGH OBSERVATIONS REPORTED AS LOW.

Model	Accuracy	Macro-F1	HIGH recall	Missed-HIGH	Mean cost
news2_rule_on_observed	0.9005	0.8006	0.8847	0.0127	0.3165
threshold_alarms	0.7352	0.6280	0.8520	0.0070	0.4461
logistic_regression	0.8677	0.7866	0.8398	0.0023	0.2893
random_forest	0.9005	0.8267	0.9013	0.0014	0.2074
gradient_boosting	0.9260	0.8324	0.9199	0.0082	0.2651

Cohort: 120,000 readings from 30,000 simulated patients, split by patient. Selection metric: `macro_f1`. Macro-F1 selects `gradient_boosting` while the asymmetric-cost criterion selects `random_forest`; the F1-optimal model carries the higher rate of HIGH observations reported as LOW. Cost weights are a stated assumption, not derived from outcome data.

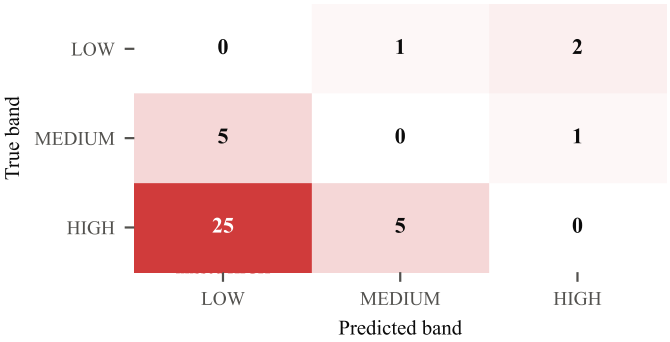


Fig. 3. The asymmetric cost matrix. Rows are the true band, columns the predicted band. The lower-left cell, a high-risk observation reported as low risk, carries 25 times the cost of a one-band over-triage. These weights encode a clinical intuition and are not derived from outcome data.

TABLE III
MACRO-F1 AGAINST SENSOR-NOISE SCALE, MEAN OVER 3 INDEPENDENT COHORT SEEDS. AT $\sigma = 0$ THE OBSERVATION EQUALS THE LATENT SIGNAL, SO THE RULE IS EXACT AND THE LEARNED MODEL CANNOT IMPROVE ON IT; THE ADVANTAGE APPEARS ONLY ONCE OBSERVATIONS ARE CORRUPTED.

σ	NEWS2 rule	gradient_boosting	Δ	95% CI
0.0	1.0000	0.9659	-0.0341	[-0.0381, -0.0300]
0.5	0.8669	0.8737	+0.0067	[+0.0030, +0.0105]
1.0	0.7976	0.8152	+0.0177	[+0.0158, +0.0195]
1.5	0.7362	0.7751	+0.0390	[+0.0277, +0.0502]
2.0	0.6819	0.7428	+0.0609	[+0.0553, +0.0665]
3.0	0.5855	0.6653	+0.0798	[+0.0644, +0.0951]

† interval includes zero; the difference is not resolved at this number of seeds.

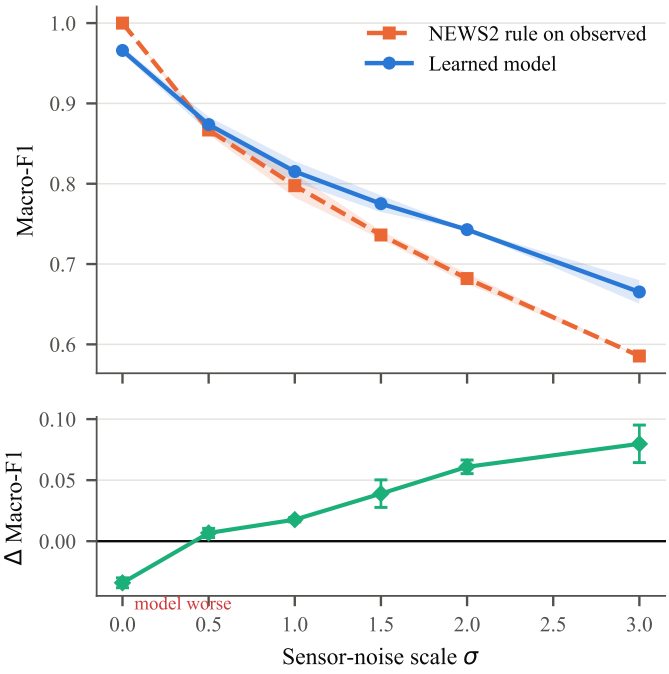


Fig. 4. Macro-F1 against sensor-noise scale (top) and the model-minus-rule difference with 95% confidence intervals over three cohort seeds (bottom). Shaded bands are 95% intervals. The difference changes sign: at $\sigma = 0$ the rule is exact and the learned model is significantly worse, and the advantage appears only once observations are corrupted.

E. Results

Table II reports all five methods on the same corrupted observations. Table III and Fig. 4 sweep the noise scale, averaging three independent cohort seeds per cell and reporting a 95% confidence interval on the model-minus-rule difference.

The negative control holds. At $\sigma = 0$ the rule is exact and the learned model is significantly worse, by -0.0341 with a 95% interval of $[-0.0381, -0.0300]$. The advantage appears once observations are corrupted and grows monotonically with noise, every interval excluding zero. This is the paper's central empirical claim, and its credibility rests on the fact that the design permits it to fail.

Fig. 5 shows where the two methods differ in kind rather than degree, and Fig. 6 breaks performance down by band. The MEDIUM band is where the aggregate figures conceal the most: it is the minority class and the boundary case, and every method does worst there.

F. Selection criteria disagree

Macro-F1 and the cost criterion select different models. The F1-optimal model attains higher aggregate scores while reporting high-risk observations as low risk 5.9 times more often than the cost-optimal alternative, for a macro-F1 difference of less than 0.006. Fig. 7 shows that the gap persists across the whole noise sweep rather than being an artifact of one operating point.

We report the disagreement and retain the pre-registered criterion. Changing it after observing results would be precisely the post-hoc selection that makes a reported number untrustworthy, and the project's own audit of someone else's dataset would sit badly beside it.

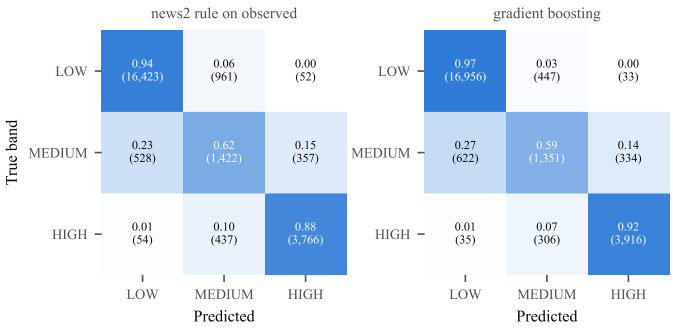


Fig. 5. Row-normalised confusion matrices for the rule baseline and the selected model, with raw counts in parentheses. Row normalisation is necessary because the bands are strongly unbalanced; raw counts would show little beyond a bright LOW cell.

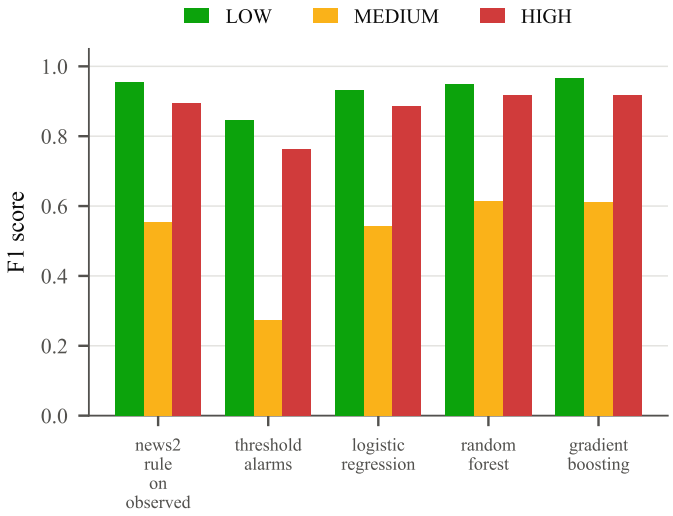


Fig. 6. Per-band F1. The MEDIUM band is the hardest for every method, which the macro average partially conceals.

G. Calibration

The system displays a confidence value, so whether that value means anything is part of the result rather than an implementation detail. Table IV and Fig. 8 give the reliability table and diagram. We expected over-confidence, since tree ensembles are known to produce voting fractions that read like probabilities without being calibrated [23]. We did not find much: expected calibration error is 0.0053 and mean confidence exceeds observed accuracy by 0.0014. The point is not that the suspicion was wrong. It is that the system previously displayed a number nobody had checked.

H. Sensor-only deployment

Two NEWS2 parameters, supplemental oxygen and consciousness, are recorded by staff rather than sensed. A deployment fed only by wearables cannot supply them, and the interface defaults them to the benign case. This is failure mode F5, and it is silent in the same way F3 is: nothing errors, the numbers simply get worse.

Table V and Fig. 9 quantify it. Serving the fully-trained model on a sensor-only feed raises the missed-HIGH rate more than three-fold. Retraining for the inputs actually present recovers most of the safety margin while barely changing macro-F1—another instance of the aggregate metric concealing the clinically relevant difference.

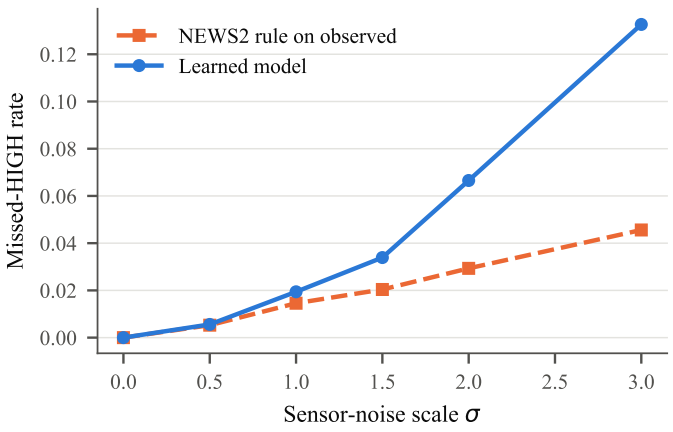


Fig. 7. Missed-HIGH rate across the noise sweep: the fraction of genuinely high-risk observations reported as low risk. This is the error the aggregate metrics hide, and it is the one a triage system exists to prevent.

TABLE IV
CALIBRATION OF THE SELECTED TRIAGE MODEL. A POSITIVE GAP MEANS STATED CONFIDENCE EXCEEDS OBSERVED ACCURACY IN THAT BIN.

Confidence bin	n	Mean conf.	Accuracy	Gap
[0.3, 0.4]	26	0.387	0.500	-0.113
[0.4, 0.5]	414	0.471	0.454	+0.016
[0.5, 0.6]	1,054	0.550	0.548	+0.001
[0.6, 0.7]	1,063	0.651	0.611	+0.039
[0.7, 0.8]	1,009	0.751	0.733	+0.018
[0.8, 0.9]	1,283	0.853	0.843	+0.010
[0.9, 1.0]	19,151	0.988	0.991	-0.002

ECE 0.0053, MCE 0.1131, Brier 0.1022, overall over-confidence +0.0014.

VI. System Architecture

A. Overview

Four components: a Python triage service exposing a stateless inference API; a Java backend owning authentication, validation, persistence, orchestration and the summarization cascade; PostgreSQL for durable telemetry and assessments; and a browser dashboard. Fig. 10 shows the arrangement and, more importantly, the boundary between what must be fast and what is permitted to be slow.

B. The decoupling rule

Telemetry ingestion performs validation and one durable write, then returns `202 Accepted`. The status code is deliberate. Returning `200` would imply the analysis was complete, which under load it is not.

Analysis is handed to a bounded executor with a caller-runs rejection policy. The bound matters more than it first appears. An unbounded queue would let a telemetry burst accumulate an ever-growing backlog of pending model calls and eventually exhaust heap, converting a load spike into an outage. With a bounded queue and caller-runs, sustained overload

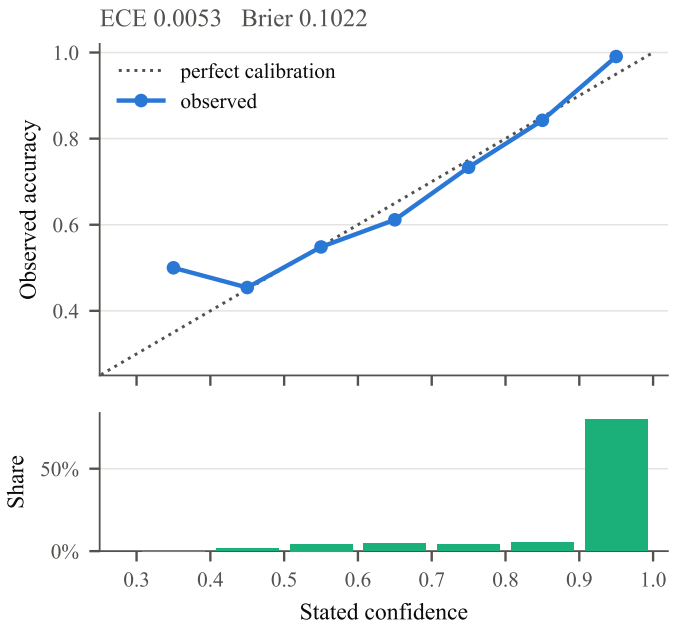


Fig. 8. Reliability diagram for the selected triage model, with bin support below. Points on the diagonal indicate that a stated confidence matches observed accuracy. Support is shown because a large gap in a sparsely populated bin is a weaker finding than the same gap in a well populated one.

TABLE V
WEARABLE-ONLY DEPLOYMENT ABLATION. SUPPLEMENTAL OXYGEN AND AVPU ARE NEWS2 PARAMETERS A WEARABLE CANNOT SUPPLY. S1: BOTH AVAILABLE (THE REPORTED SETTING). S2: MODEL TRAINED WITH THEM BUT SERVED WITHOUT. S3: RETRAINED FOR THE REDUCED INPUT SET.

Scenario	Macro-F1	HIGH recall	Missed-HIGH
S1 full observation	0.8214	0.8891	0.0051
S2 silent mismatch	0.7783	0.7803	0.0167
S3 retrained	0.7744	0.8448	0.0109

Serving the fully-trained model on a wearable-only feed costs +0.0431 macro-F1 and raises missed-HIGH by +0.0116.

instead pushes back on the ingestion thread: ingestion slows, nothing is dropped, and nothing grows without limit. This is requirement R2, and it follows the tail-latency reasoning of [32]—an occasionally slow dependency damages a coupled system far more than its mean latency suggests.

Fig. 11 shows the timing for a single reading, with the point of acknowledgement marked.

C. Degraded operation

When the triage service is unreachable or exceeds its timeout, the assessment is recorded with the band marked unavailable, the breakdown absent, and a flag indicating the model did not contribute. It is not inferred, not carried forward from a previous reading, and not defaulted to LOW. The summary is still produced, from the numbers, by the deterministic tier. Fig. 12 shows the two states and what they share.

D. Persistence and provenance

Fig. 13 gives the schema. Derived features are persisted rather than recomputed on read, so the dashboard and the

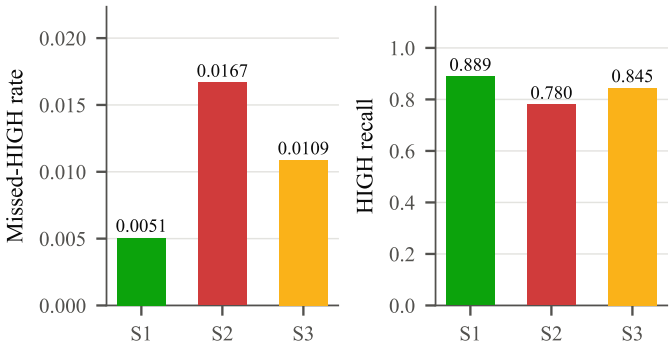


Fig. 9. Sensor-only deployment. Serving the fully-trained model against a feed lacking supplemental-oxygen and consciousness observations (S2) raises the rate of high-risk observations reported as low risk more than three-fold. Retraining for the inputs actually present (S3) recovers most of the safety margin while barely changing macro-F1.

benchmarks cannot drift from the definitions used at write time. More importantly, four provenance columns are stored with every assessment: which tier answered, its latency, whether triage was reachable, and whether verification passed. This is requirement R6, and it is what allows the resilience claims in Section IX to be measured from stored data rather than scraped from logs.

VII. FAITHFULNESS-VERIFIED SUMMARIZATION

A. Why prompting is not enough

The prompt instructs the model not to invent findings. Prompts are not enforcement. The summarization literature is unambiguous that instruction following does not prevent hallucination [27], [28], and in a clinical context the specific failure that matters is a fabricated number, because a number looks checkable and therefore invites trust.

B. Definition

Let R be a summarization request carrying the latest reading, the derived features, the NEWS2 result, and trend deltas over a bounded window. Define the *supported set* $\mathcal{A}(R)$ as the union of all values the request legitimately contains, together with their rounded forms at integer, one-decimal and two-decimal precision, and the reference-range constants the deterministic tier itself quotes.

A candidate summary T is *faithful* with respect to R when

$$\forall v \in \mathrm{num}(T) \;\; \exists\, \alpha \in \mathcal{A}(R) : \; |v - \alpha| \le \tau \qquad (6)$$

and T contains no term from a list of condition names, where $\mathrm{num}(T)$ extracts numeric literals from T after removing measurement names that end in a digit, and τ is an absolute tolerance.

C. Choosing the tolerance

The tolerance is the whole design. Set it loose and the check is decorative; set it tight and legitimate rounding is rejected.

Our first implementation used $\tau = 1.0$, which seemed conservative. It was not. A test summary inventing "lactate 4.7 mmol/L" passed, because the patient's NEWS2 total was 5 and

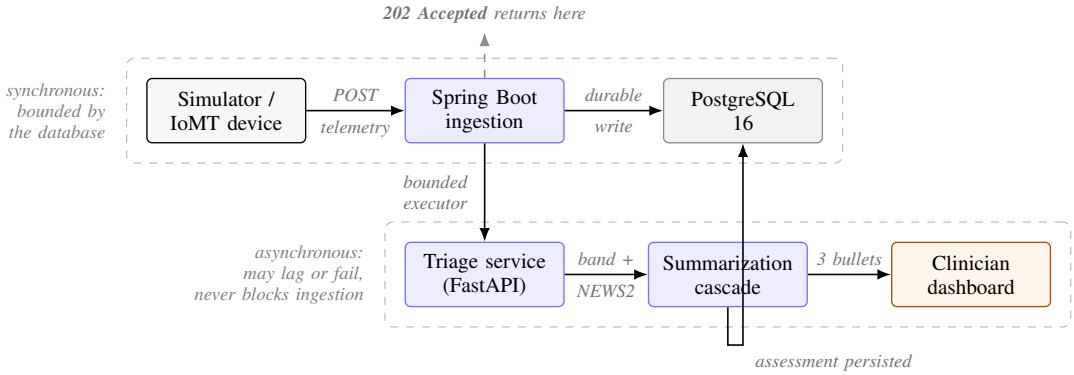


Fig. 10. Architecture and the synchronous/asynchronous boundary. Ingestion performs one durable write and returns `202 Accepted`; triage and summarization proceed on a bounded executor behind it. Ingestion throughput is therefore bounded by the database, never by whichever language model is currently answering.

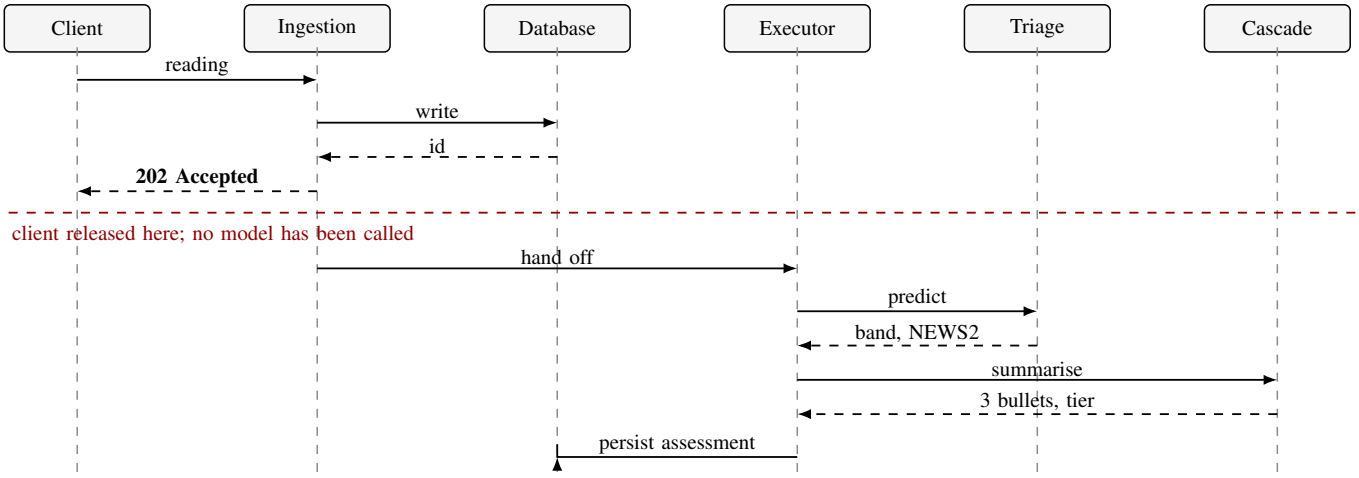


Fig. 11. Timing for one reading. The client is released after the durable write and before any model is consulted, which is what makes acknowledgement latency independent of triage and summarization latency (requirement R1).

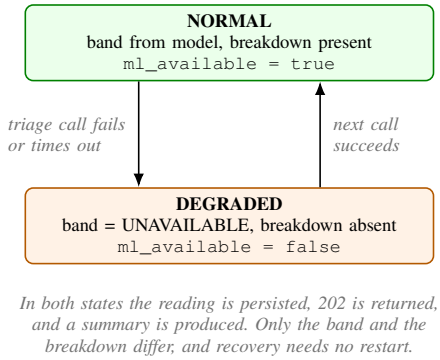


Fig. 12. Degraded-mode state machine. Ingestion, persistence and summarization are available in both states; only the risk band and the score breakdown differ. Recovery requires no restart.

$|4.7 - 5| = 0.3 < \tau$. A fabricated clinical value had matched an unrelated score. The lesson generalises: a tolerance tuned against large values is a hole at small ones. One unit on a heart rate of 118 is 0.8%; on a score of 5 it is 20%.

The fix was to enumerate the legitimate rounded forms explicitly in $\mathcal{A}(R)$ and then reduce τ to 0.05, which now absorbs only floating-point representation error. That case is retained as a regression test.

D. Cascade

Algorithm 1 gives the procedure, and Fig. 14 the flow. Two properties follow directly. Failing verification is treated as a tier failure rather than as a result, so a fabricating tier is skipped rather than surfaced (R5). And the terminal tier has no external dependency and is verified in test to satisfy Equation 6 by construction, so the loop always terminates with a summary (R4).

E. What this does not do

The check is necessary, not sufficient, and we want to be explicit about the gap. It catches invented values and named conditions. It cannot detect a summary that is fluent, numerically correct, and clinically wrong—a correct recitation of the numbers paired with a misleading emphasis would pass. Nor is it a substitute for clinician review. Entailment-based approaches [28] would extend the check beyond numerals, and we regard that as the natural next step rather than as a solved problem.

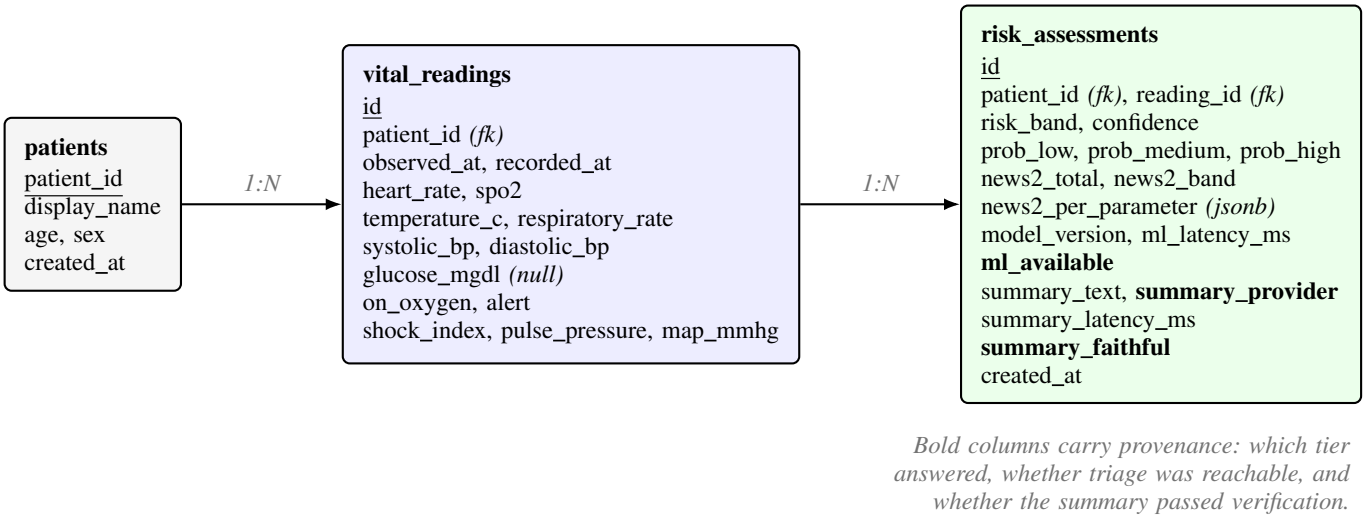


Fig. 13. Persistence schema. Provenance columns, shown in bold, record which summarization tier answered, whether the triage service was reachable, and whether the generated text passed verification.

Algorithm 1 Verification-gated summarization cascade

Input: request R; ordered tiers $g_1, \ldots, g_n$ with g_n deterministic
Output: summary T, tier identifier, verification result
for $i = 1$ **to** n **do**
  if g_i is not configured or not reachable **then**
    continue
  end if
  $T \leftarrow g_i(R)$ {never throws; may return $\emptyset$}
  if $T = \emptyset$ **or** T has fewer than three bullets **then**
    continue {F2: tier failure}
  end if
  $\phi \leftarrow \textsc{Faithful}(T, R)$ by Eq. 6
  if $\neg\phi$ **and** $i < n$ **then**
    continue {F3: treated as tier failure}
  end if
  return (T, i, ϕ)
end for

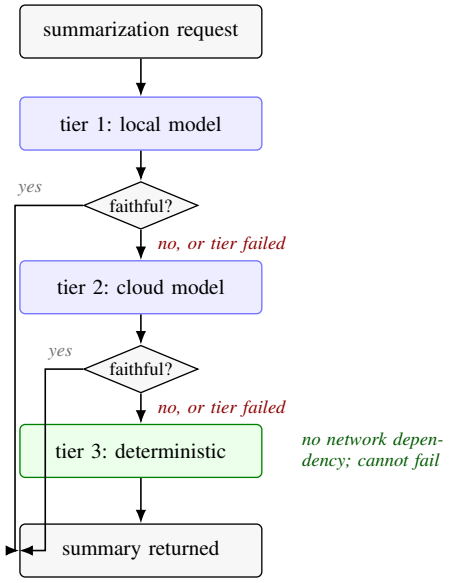


Fig. 14. Summarization cascade. A tier that fails, times out, or returns output failing the faithfulness check is skipped. The deterministic tier has no external dependency and therefore always produces a summary, which is the property the availability claim rests on.

VIII. Implementation

The triage service is Python with scikit-learn [22], serving one persisted pipeline that contains the imputer, the scaler and the estimator in a single artifact. Shipping these separately is a well-known route to silent train/serve skew [33]; one artifact makes the skew impossible. A model card recording the artifact hash, the selection metric, the headline metrics and the intended use is stored alongside it.

The backend is Java with Spring Boot. Persistence uses explicit SQL through `JdbcClient` with Flyway migrations rather than an object-relational mapper, because the headline claim concerns latency and hidden lazy-loading behaviour would undermine the measurements. The executor is configured with core 4, maximum 8, queue 500, and a caller-runs rejection policy. Outbound calls to the triage service carry a 2 s connect and 5 s read timeout; without explicit timeouts a stalled dependency would hold an analysis thread indefinitely and eventually exhaust the pool.

Both language model tiers are optional beans. With neither configured the application starts normally and the deterministic tier answers every request. This is not an incidental property: it is what allows the system to be deployed, tested and reproduced with no API keys and no local model weights.

The dashboard is served by the backend as static assets with no build step. It surfaces the ward sorted by acuity, the per-parameter NEWS2 breakdown beside the model band, and—deliberately—which tier produced each summary and whether it passed verification.

TABLE VI
TRIAGE INFERENCE LATENCY, 1000 SINGLE-OBSERVATION PREDICTIONS. IN-PROCESS MODEL CALL ONLY: EXCLUDES HTTP, SERIALISATION AND NETWORK TRANSIT.

Statistic	Milliseconds
Mean	2.758
p50	2.684
p95	3.719
p99	4.297
Maximum	4.687

Measured on one developer workstation. Not a service-level objective.

IX. EVALUATION

A. Triage latency

Triage is the only model-dependent step in the synchronous-facing path, so its cost bounds how much of the ingestion budget analysis can consume. Table VI reports the in-process model call over 1000 single-observation predictions. The distribution is tight and the tail is short, which is what permits analysis to be scheduled per reading rather than batched.

B. R1: decoupling under a stalled dependency

With the triage service artificially stalled for 3 s, ingestion returns in under 1 s. A synchronous implementation cannot pass this test, which is why it is the load-bearing test of the architectural claim.

This test also taught us something about measurement. The first version failed, and the tempting conclusion was that the asynchronous dispatch was broken. It was not. The test was performing an authentication round-trip inside the timed region, and on first call also paying Spring Security filter initialisation and JIT warm-up. Hoisting both out of the measurement and tightening the bound from 1500 ms to 1000 ms produced a test that passes comfortably and measures what it claims. Changing working architecture to satisfy a broken measurement would have been the worse outcome by some distance.

C. R2: throughput and saturation

Offered load was generated open-loop. This matters: a closed-loop generator that waits for each reply before sending the next silently throttles itself to whatever the server can manage, and therefore cannot detect saturation at all. With a fixed schedule, a server that falls behind manifests as rising latency.

At 10, 50 and 100 requests per second the system tracks the offered rate with p95 ingestion latency between 27 and 31 ms. At an offered 250 it saturates, achieving approximately 159 requests per second with p95 latency rising by two orders of magnitude. The asynchronous analysis backlog drained completely at every rate, which is the behaviour requirement R2 describes: the system slows rather than failing or accumulating unbounded work. Fig. 15 shows both panels; achieved rate and latency are plotted separately rather than on twin axes, since they share no scale.

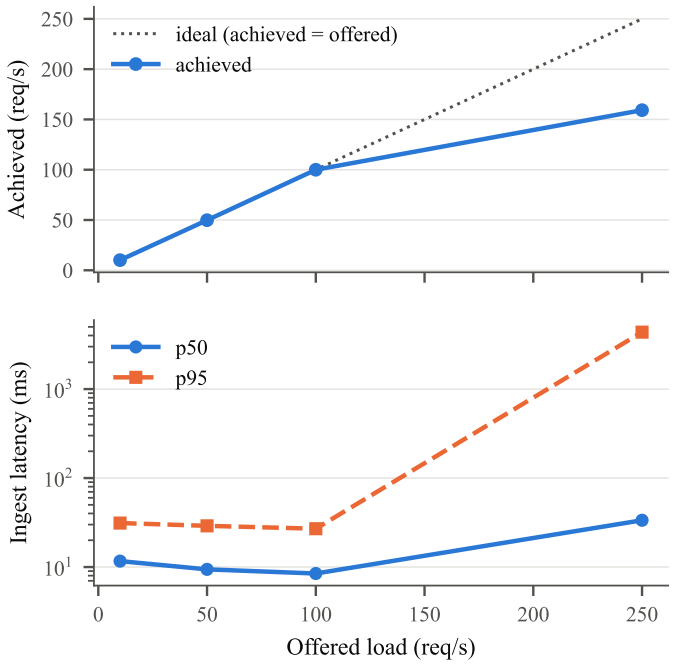


Fig. 15. Ingestion under open-loop offered load. Achieved rate tracks offered load to 100 requests per second and then departs from it; p95 latency rises by two orders of magnitude at the saturation point. Latency uses a logarithmic axis.

D. R3, R4, R6: resilience

Three conditions were exercised by stopping and restarting containers rather than by mocking. With all services running, every reading produced an assessment. With the triage service stopped, ingestion and summarization both remained at 100%, assessments were recorded with the model marked unavailable and the band reported as unavailable rather than guessed, and summaries were produced by the deterministic tier. On restart the system resumed using the model without a backend restart.

The recovery condition initially reported a false failure, which is worth recording. The readiness check polled the triage container's own health endpoint from the host, which answers the question "is the process listening" rather than the question that matters, "can the backend obtain a prediction". Replacing it with an end-to-end probe—submit a reading, wait for an assessment with the model flag set—produced the correct result. Readiness is properly defined from the consumer's point of view.

E. R5: faithfulness enforcement

Under test, a candidate summary inventing a lactate value is rejected and the cascade falls through; the deterministic tier is verified to pass the same check on normal, severely deranged, bradycardic and hypertensive inputs. In live operation every summary recorded so far has been produced by the deterministic tier and has passed, because no language model has been enabled. This is a real limitation and is discussed in Section XII.

TABLE VII
NOTIFICATION EVENT COUNTS OVER 4,800 SIMULATED READINGS. A EMITS ONE EVENT PER BREACHING PARAMETER PER READING; B ONE PER READING WHOSE BAND IS NOT LOW; C ONE PER UPWARD BAND TRANSITION PER PATIENT. THIS IS EVENT ARITHMETIC ONLY.

Policy	Events	Per 100
A. Per-parameter thresholds	4,306	89.71
B. Aggregate band	1,262	26.29
C. Band transition	419	8.73

These counts are not a measurement of alert fatigue, clinician workload, or patient benefit. Policy C emits fewer notifications by construction; whether that is clinically safe is untested here.

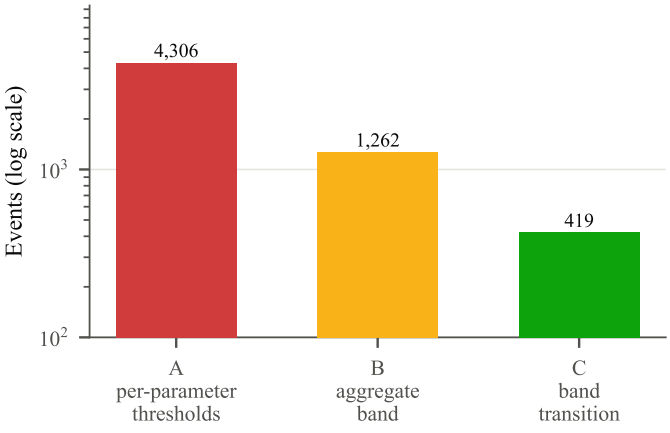


Fig. 16. Notification volume under three policies over the same simulated stream, on a logarithmic scale. These are event counts only.

F. Notification event counts

We stress that this is arithmetic over a simulated stream. It is not a measurement of alert fatigue, clinician workload, or patient benefit, and it must not be read as one. A policy emitting fewer notifications is not thereby safer; fewer notifications can mean missed deterioration. Establishing clinical benefit would require the kind of prospective study reported in [9], which is well beyond anything demonstrated here.

G. Test suite

The implementation carries 151 automated tests: 104 in Python covering NEWS2 boundary conditions at every threshold, the zero-noise generator identity, patient-level split integrity, the inference API contract, and the clinical and calibration metrics; and 47 in Java covering token handling, every combination of cascade tier failure, the faithfulness guard, and end-to-end integration against a real PostgreSQL instance under Testcontainers.

X. LIVE LANGUAGE MODEL EVALUATION

This section is reserved for evaluation of the live summarization tiers and is deliberately unpopulated in the present version. The cascade, its failure paths and the faithfulness guard are implemented and tested, but no language model has been executed: every summary reported in this paper was produced by the deterministic tier. The faithfulness guard has therefore been shown to reject fabricated output only under test, and its behaviour against real model output is unmeasured.

We regard reporting this as an unmeasured gap as preferable to omitting the section. The availability argument of Section VI rests on tiers whose generative behaviour has not yet been characterized, and that limit should be visible where a reader looks for those results.

XI. DISCUSSION

A. On auditing before modeling

The Track A finding was not subtle and required no sophisticated tooling. A depth-3 tree exposes it in seconds. It survives in the literature because the standard workflow never asks the question. Given that leakage has been documented across hundreds of papers in seventeen disciplines [17], we suggest that a label-provenance check should be as routine as a train/test split, and that it is cheap enough that there is no good reason to skip it.

B. On negative controls

The $\sigma = 0$ result, where the learned model is significantly worse than the rule, is the most informative number in Table III. A design in which the model wins everywhere would be indistinguishable from a design that leaks. This connects to the generalizability critique [15]: a result that cannot fail under any condition is not evidence about the conditions it claims to describe.

C. On aggregate metrics in clinical triage

Macro-F1 selected the worse model twice in this work, under different circumstances: once between estimator families, once between input configurations. In both cases the cost-relevant difference was invisible in the aggregate score. Where error costs are asymmetric—and in triage they always are—directional error rates belong alongside aggregate metrics, not in an appendix. The cost-sensitive framing is long established [24], [26]; the observation here is that it is still routinely omitted from reported results.

D. On architecture as a safety property

It is tempting to treat the decoupling as an engineering convenience. It is not. Under F1 a coupled system has two options, both bad: block ingestion, or fabricate a band. The architecture removes the dilemma by making the degraded state representable and durable. Much of what this paper calls resilience is really just refusing to let an unavailable component be silently replaced by a guess.

XII. LIMITATIONS

1) The Track B cohort is simulated. Results are software evaluation, not clinical validation, and the corruption model is a stated assumption rather than a measured device characterisation.
2) No language model has been run live. Section X is unpopulated and the faithfulness guard is unvalidated against real generative output.

3) Faithfulness checking cannot detect fluent, numerically correct, clinically incorrect interpretation.
4) The asymmetric cost weights encode intuition, not outcome data.
5) NEWS2 is validated for acutely ill adults in hospital [1], [2]; applying it to simulated continuous wearable telemetry is outside its validation scope, and the methodological critique of early warning scores [4] applies.
6) A sensor-only deployment loses two of seven NEWS2 parameters, with the measured consequences in Table V.
7) Security is prototype grade: a single demo credential, no refresh or revocation, no external identity provider, no TLS in the bundled stack.
8) Throughput figures characterize one workstation and are not service level objectives.
9) The evaluation uses a single simulated cohort generator. Dataset shift between a generator and reality is exactly the hazard described in [14], and nothing here measures it.

XIII. Ethics and Safety

The system is a research prototype and not a medical device. It produces software triage categories, not diagnoses, and every API response and every screen carries that statement. No real patient data was used at any stage; all cohorts are simulated and the audited public dataset contains no identifiable information.

Enabling the cloud summarization tier transmits telemetry to a third party, which is acceptable for synthetic data and would require a data processing agreement otherwise. Deployment on real data would require the hardening enumerated under Limitations, and a determination of regulatory status under applicable Clinical Decision Support guidance [36]. Risk management for the generative component would follow the NIST generative AI profile [35].

XIV. Reproducibility

All results are regenerated from a clean checkout by a single script. Each experiment writes a JSON artifact recording its seeds, configuration and environment; every table and figure in this paper is generated from those artifacts, and no number is entered by hand. The audited dataset is not redistributed; a resolver script locates it and records its SHA-256 so a result can be tied to a specific file. Track B requires no external data.

XV. Conclusion

VitalSync demonstrates that continuous telemetry ingestion and low-latency risk triage can be isolated from generative summarization, remaining fully available when every language model tier is unreachable, and that generated clinical text can be held to a mechanical grounding check that governs fallback rather than merely annotating output.

Along the way we found that the public dataset the work was premised on carries a label that is a deterministic rule over two variables, and we report that as a caution about label provenance. Separating latent physiology from sensor observation gives a falsifiable comparison in which a learned model outperforms a threshold rule only under observation noise, and demonstrably fails to when there is none. Finally, we show that aggregate classification metrics can select triage models that are materially worse under asymmetric clinical cost.

Evaluating the generative tiers themselves remains open, and is marked as such. So is the obvious next step for the verification gate: extending it from numeric entailment to textual entailment, so that it catches claims the current check cannot see.

Appendix A
NEWS2 Parameter Scoring

Table VIII reproduces the per-parameter scoring used as the labelling function, following the Royal College of Physicians specification [1]. The aggregate is the sum of the seven parameter scores. The escalation band is HIGH when the aggregate is at least 7, MEDIUM when the aggregate is 5 or 6 *or* any single parameter scores 3, and LOW otherwise. The single-parameter clause is the part most often omitted in reimplementations, and it is the reason the band is not a monotone function of the aggregate alone.

TABLE VIII
NEWS2 per-parameter scoring (Scale 1).

Parameter	Range	Score
Respiration rate (min^{-1})	≤ 8	3
	9–11	1
	12–20	0
	21–24	2
	≥ 25	3
SpO_2 (%)	≤ 91	3
	92–93	2
	94–95	1
	≥ 96	0
Supplemental oxygen	yes / air	2 / 0
Systolic BP (mmHg)	≤ 90	3
	91–100	2
	101–110	1
	111–219	0
	≥ 220	3
Pulse (min^{-1})	≤ 40	3
	41–50	1
	51–90	0
	91–110	1
	111–130	2
	≥ 131	3
Consciousness	alert / CVPU	0 / 3
Temperature (°C)	≤ 35.0	3
	35.1–36.0	1
	36.1–38.0	0
	38.1–39.0	1
	≥ 39.1	2

Appendix B
Sensor Corruption Parameters

Per-channel corruption at noise scale $\sigma = 1$: Gaussian standard deviations of 2.5 bpm (heart rate), 1.5% (oxygen saturation), 1.5 min^{-1} (respiratory rate), 0.15 °C (temperature),

5 mmHg (systolic) and 4 mmHg (diastolic). Motion artifact is applied to 1.5% of heart-rate readings, adding 15–40 bpm. Perfusion dropout is applied to 2% of oxygen-saturation readings, subtracting 3–8 percentage points. Whole-channel dropout occurs at 3% per channel and is median-imputed at inference. Display quantization is applied to integers for heart rate, respiratory rate and both pressures, and to one decimal for oxygen saturation and temperature.

Quantization is part of the corruption model and is disabled at $\sigma = 0$. Without that, rounding alone pushes values across NEWS2 thresholds and the zero-noise identity fails at approximately 0.984 rather than exactly 1, which is how the issue was originally found.

APPENDIX C
COST MATRIX

The asymmetric cost matrix C, indexed $C[\text{true}][\text{predicted}]$ over $(\text{LOW}, \text{MEDIUM}, \text{HIGH})$:

$$C = \begin{bmatrix} 0 & 1 & 2 \\ 5 & 0 & 1 \\ 25 & 5 & 0 \end{bmatrix} \tag{7}$$

Under-triage is penalised more heavily than over-triage, and missing a high-risk observation entirely is penalised an order of magnitude beyond a one-band slip. These weights are an assumption. They encode the view that escalating unnecessarily costs a review while failing to escalate can cost a patient, but they are not derived from outcome data and a reader who disagrees should substitute their own and rerun.

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